

Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Simultaneous Localization and Mapping (SLAM)

Dec 17, 2025

Welcome to the lecture!

Prof. Dr.-Ing. Björn Giesler

Technische Hochschule Ingolstadt

AIMotion, AI and Sensor Data Fusion

Email: bjoern.giesler@thi.de

- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

Sensor Networks and Sensor Fusion

- **Previous lectures:**
  - Mathematical details of Particle Filters
  - Joint Probability Data Association Filters
  - JPDAs and Particle Filters

- **This week:**
  - Simultaneous Localization and Mapping
    - Introduction
    - EKF SLAM
    - Loop Closure
    - Particle Filter / Rao-Blackwellized Particle Filter SLAM
    - "Hidden Robot" problem
- **After the Christmas Break:**
  - Bit more SLAM
  - (DNN-based) early fusion approaches.

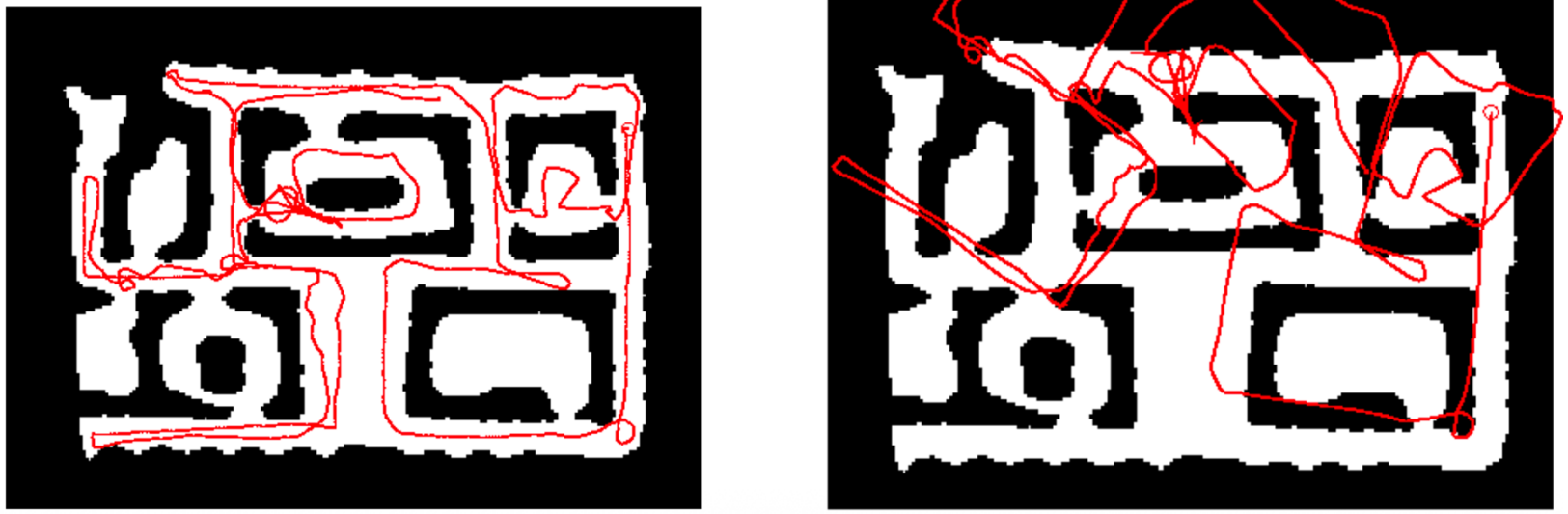


Merry Robot Christmas!

Robot Motion and Odometry

Robot Motion

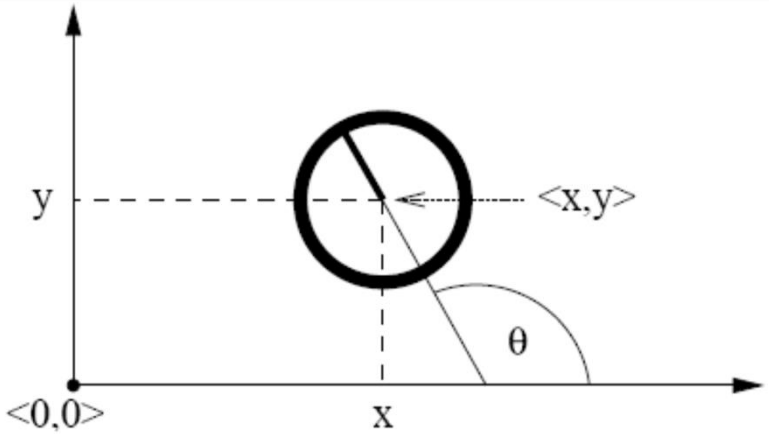
- **Robot motion is inherently uncertain. How can we model this uncertainty?**

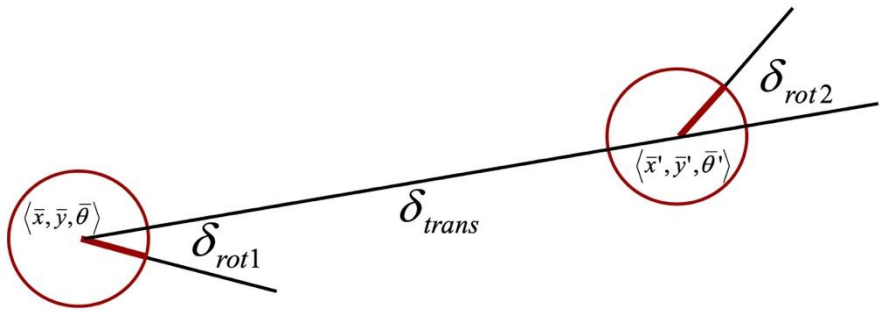


Robot Motion

A freely moving robot in 3D is described with six parameters: $(x, y, z, \theta_{\text{yaw}}, \theta_{\text{pitch}}, \theta_{\text{roll}})^\top$.

For simplicity, we consider 2-wheeled robots operating on a planar surface: $(x, y, \theta)^\top$.





2-wheeled robots can rotate, drive straight lines, or drive curves which can be hard to model.

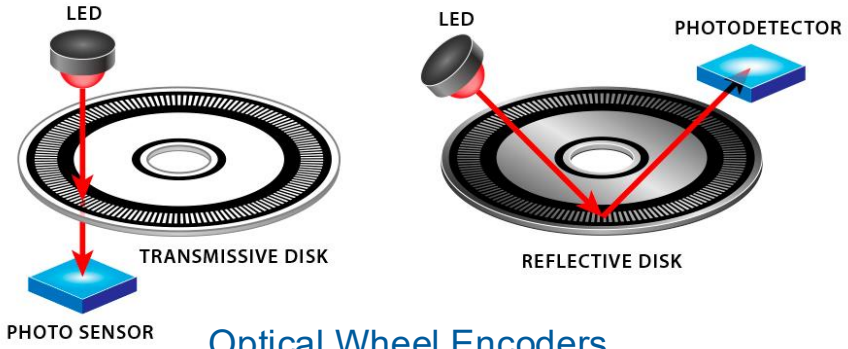
For simplicity, we linearize motion into three steps:

$$\delta_{\text{rot1}} = \arctan \frac{\hat{y}' - \hat{y}}{\hat{x}' - \hat{x}} - \hat{\theta}$$

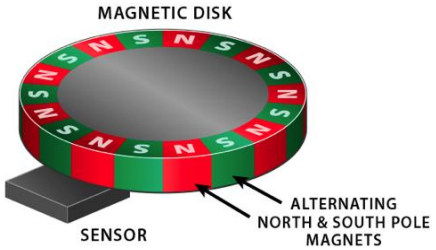
$$\delta_{\text{trans}} = \sqrt{(\hat{x}' - \hat{x})^2 + (\hat{y}' - \hat{y})^2}$$

$$\delta_{\text{rot2}} = \hat{\theta}' - \hat{\theta} - \delta_{\text{rot1}}$$

Odometry

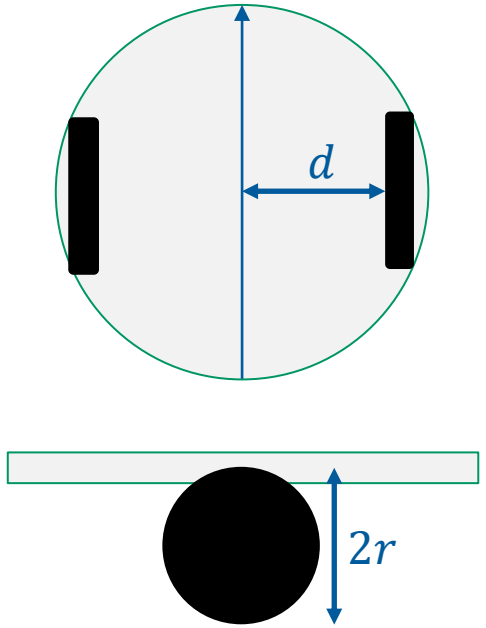


Optical Wheel Encoders



Magnetic Wheel Encoder

Robot top and side view



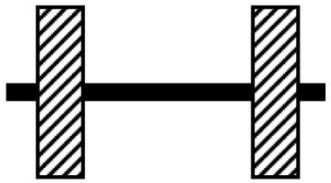
$$\delta_l = \frac{p_l}{n} \cdot 2\pi r$$

$$\delta_r = \frac{p_r}{n} \cdot 2\pi r$$

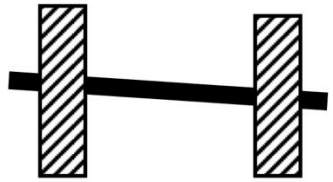
$$\delta_{\text{trans}} = \frac{\delta_l + \delta_r}{2}$$

$$\delta_{\text{rot1}} = \delta_{\text{rot2}} = \arctan \frac{\delta_l - \delta_r}{2d}$$

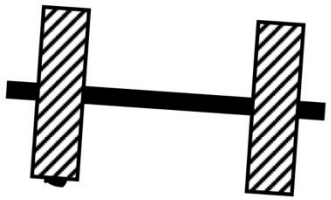
Motion Errors in Wheeled Robots



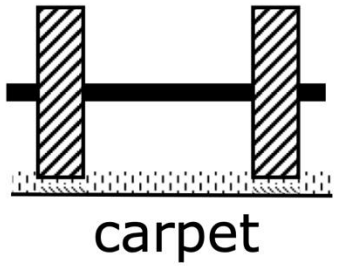
ideal case



different wheel diameters



bump



carpet

...and many more.

These may be slight, but they aggregate over time and distance driven.

They are dependent on distance driven, so we model them using four constants $\alpha_1, \alpha_2, \alpha_3, \alpha_4$:

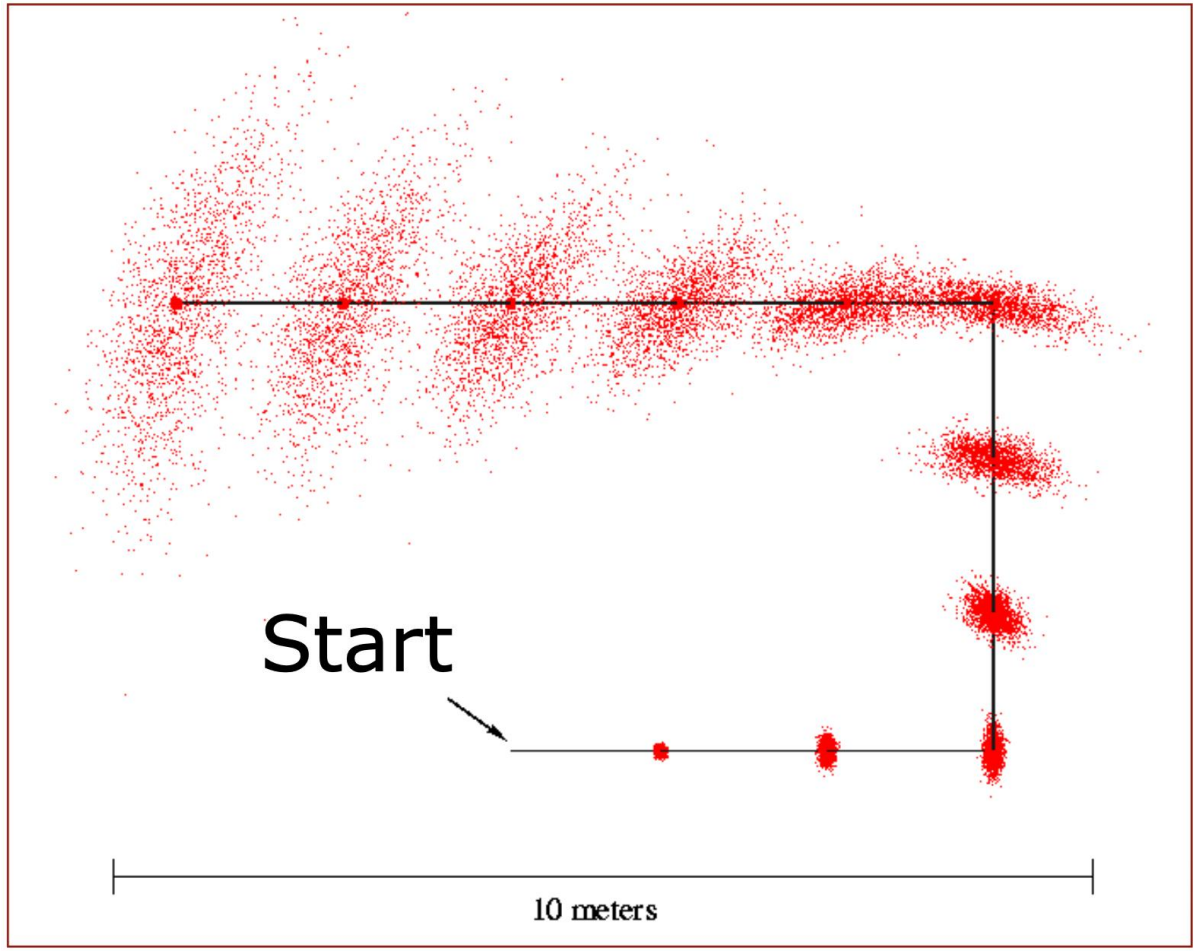
$$\hat{\delta}_{\text{rot1}} = \delta_{\text{rot1}} + \epsilon_{\alpha_1|\delta_{rot1}|+\alpha_2|\delta_{trans}|}$$

$$\hat{\delta}_{\text{trans}} = \delta_{\text{trans}} + \epsilon_{\alpha_3|\delta_{trans}|+\alpha_4|\delta_{rot1}+\delta_{rot2}|}$$

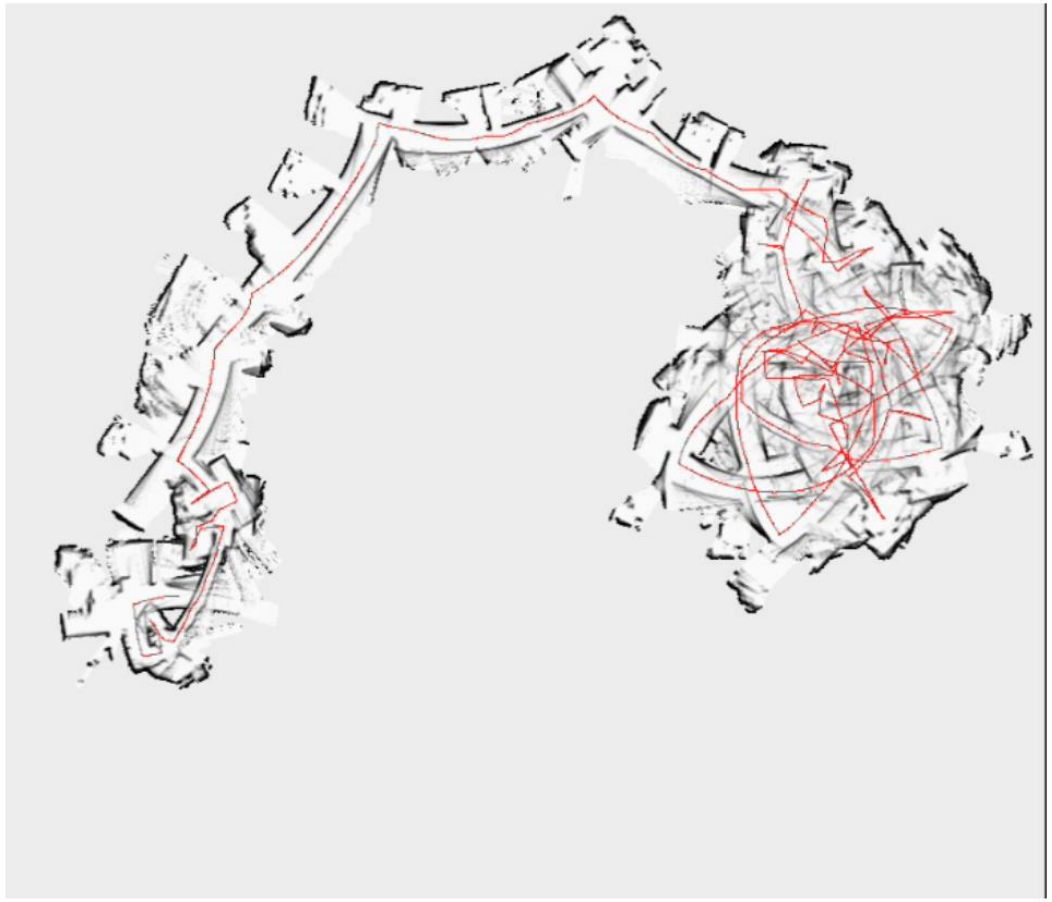
$$\hat{\delta}_{\text{rot2}} = \delta_{\text{rot2}} + \epsilon_{\alpha_1|\delta_{rot2}|+\alpha_2|\delta_{trans}|}$$

where $\epsilon_\sigma \sim N(0, \sigma)$.

Sampling from our Motion Model



Mapping with Raw Odometry



Picture: Dirk Hähnel

SLAM

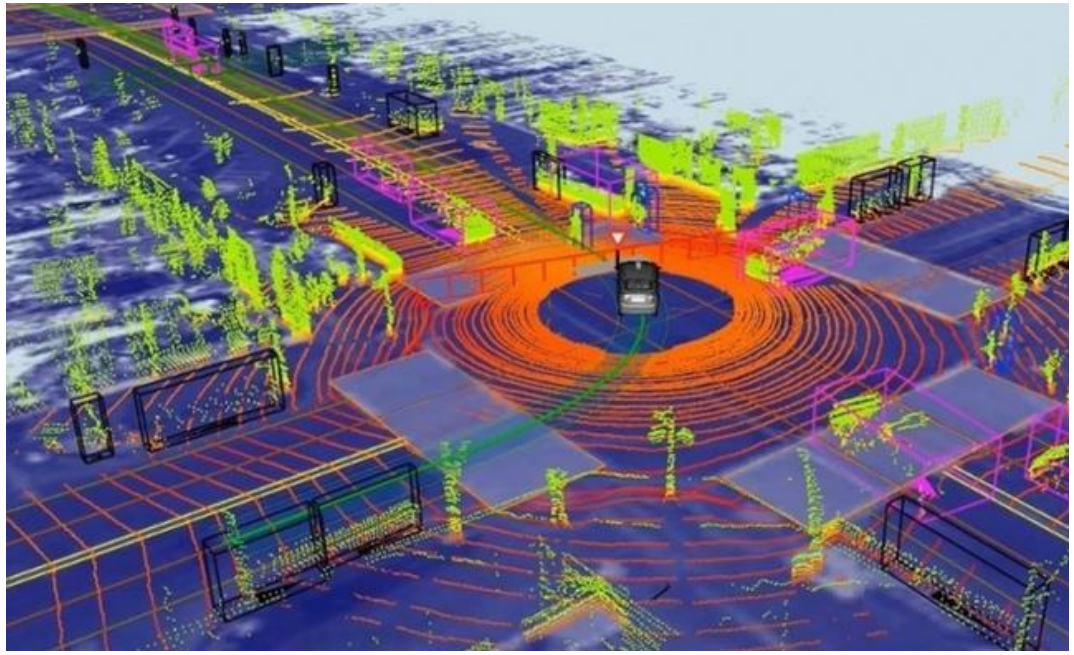
Simultaneous Localization and Mapping

Simultaneous Localization and Mapping (SLAM)

- Process of a robotic system to build a map and localize itself within this map at the same time
- Challenge: Hen-Egg Problem
  - Localizing itself requires a map
  - Mapping requires knowledge of the location
- Estimate robot state $\boldsymbol{x}_k$ and landmark $\boldsymbol{m} = \{\boldsymbol{m}_1, \ldots, \boldsymbol{m}_N\}$
- In probabilistic terms, the SLAM problem tries to derive the probability density

$$p(\boldsymbol{x}_k, \boldsymbol{m} | \boldsymbol{y}_{1:k}, \mathbf{u}_{1:k})$$

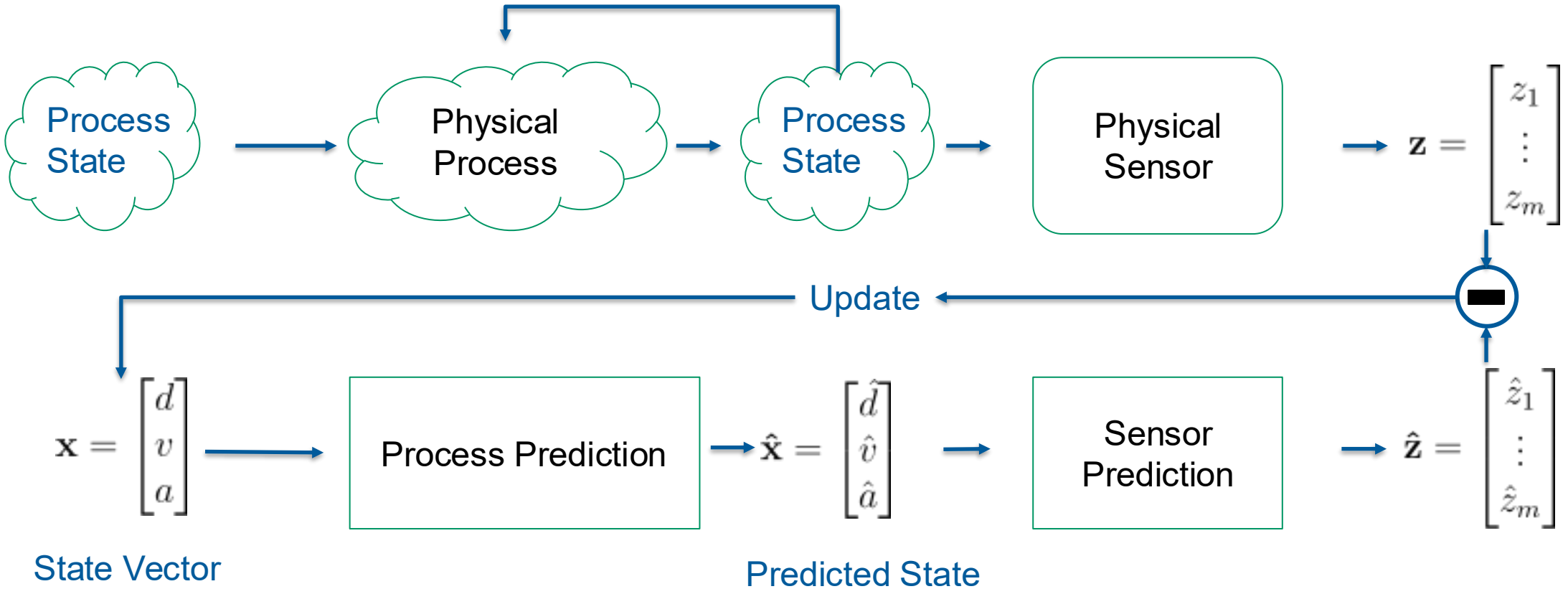
based on the history of measurements $\boldsymbol{y}_{1:k}$ and inputs $\boldsymbol{u}_{1:k}$

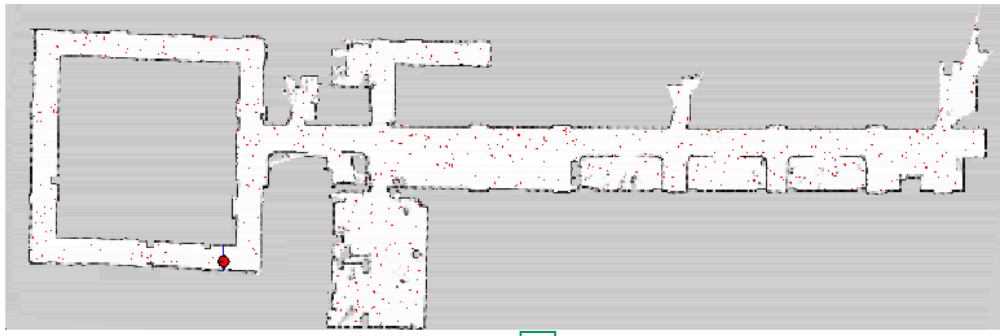


Lidar point cloud data

These slides owe a lot to Cyril Stachniss's SLAM course at Uni Freiburg

Map


Process State
Physical Process
Process State
Physical Sensor
$\mathbf{z} = \begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix}$
Update
$\mathbf{x} = \begin{bmatrix} d \\ v \\ a \end{bmatrix}$
Process Prediction
$\hat{\mathbf{x}} = \begin{bmatrix} \hat{d} \\ \hat{v} \\ \hat{a} \end{bmatrix}$
Sensor Prediction
$\hat{\mathbf{z}} = \begin{bmatrix} \hat{z}_1 \\ \vdots \\ \hat{z}_m \end{bmatrix}$
State Vector
Predicted State



Map

Process State

Physical Process

Process State

What landmarks should we see and where?

$$\mathbf{z} = \begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix}$$

What landmarks do we actually see and where?

Update

$$\mathbf{x} = \begin{bmatrix} d \\ v \\ a \end{bmatrix}$$

Process Prediction

$$\hat{\mathbf{x}} = \begin{bmatrix} \hat{d} \\ \hat{v} \\ \hat{a} \end{bmatrix}$$

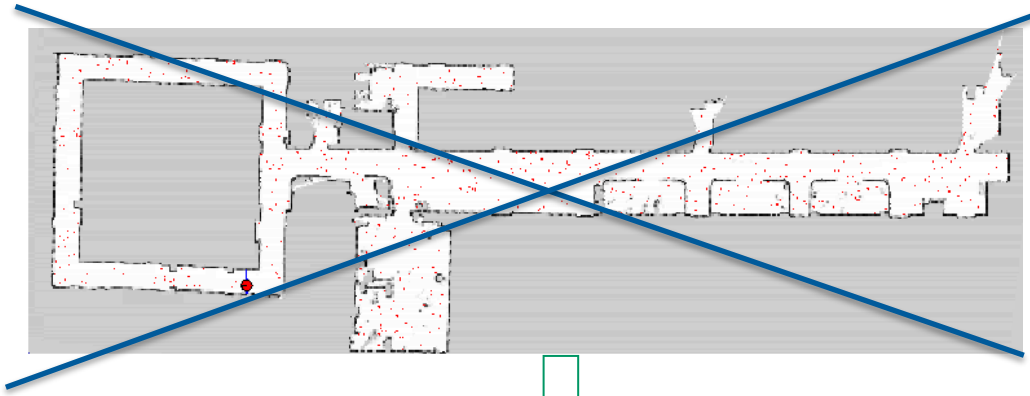
➔ Where are we?

➔ Update model

$$\hat{\mathbf{z}} = \begin{bmatrix} \hat{z}_1 \\ \vdots \\ \hat{z}_m \end{bmatrix}$$

State Vector

Predicted State



Map

Process State

Physical Process

Process State

Physical Sensor

$\mathbf{z} = \begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix}$

???

Update

$\mathbf{x} = \begin{bmatrix} d \\ v \\ a \end{bmatrix}$

Process Prediction

$\hat{\mathbf{x}} = \begin{bmatrix} \hat{d} \\ \hat{v} \\ \hat{a} \end{bmatrix}$

Sensor Prediction

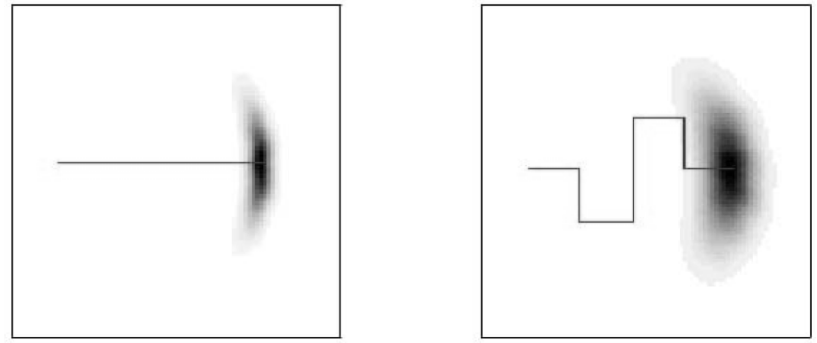
$\hat{\mathbf{z}} = \begin{bmatrix} \hat{z}_1 \\ \vdots \\ \hat{z}_m \end{bmatrix}$

State Vector

Predicted State

SLAM – Simultaneous Localization and Mapping

- **If we have perfect knowledge of our position, but can only make noisy observations of our surroundings, we can still store landmarks in a map. Filters like [EU]KF help us reduce the uncertainty on the landmarks.**
- **If we have perfect knowledge of landmark positions in a map, we can observe them and correct the noisy observation of our position. Filters like [EU]KF help us reduce the uncertainty on the egomotion.**
- **What can we do if we have**
  - rising uncertainty in our own position due to noisy egomotion sensors,
  - noisy landmark observations?
- **This is a fundamental problem for autonomous robots navigating changing environments:**
  - Autonomous cars
  - Humanoid robots
  - Vacuum cleaners, lawn mowers
  - Aerial drones, planetary rovers, mine robots, underwater drones, ...

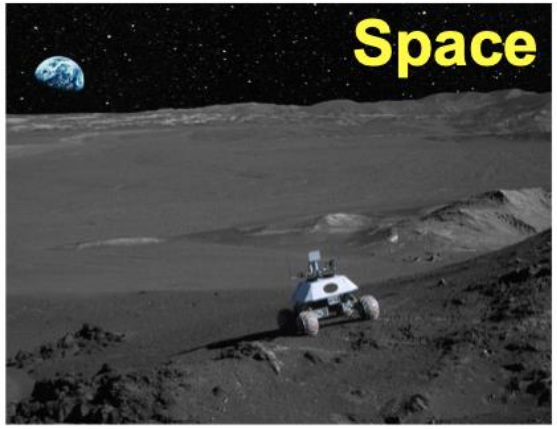


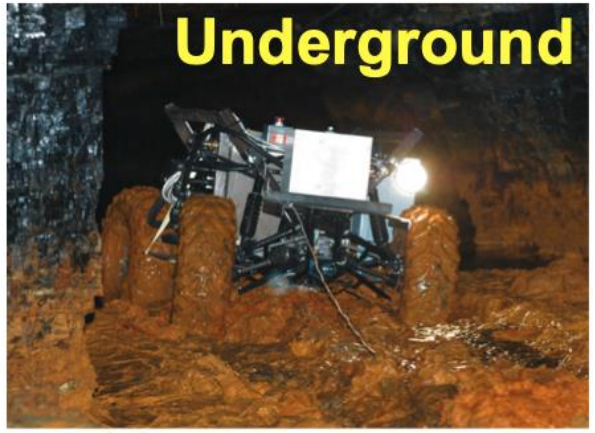


SLAM Applications

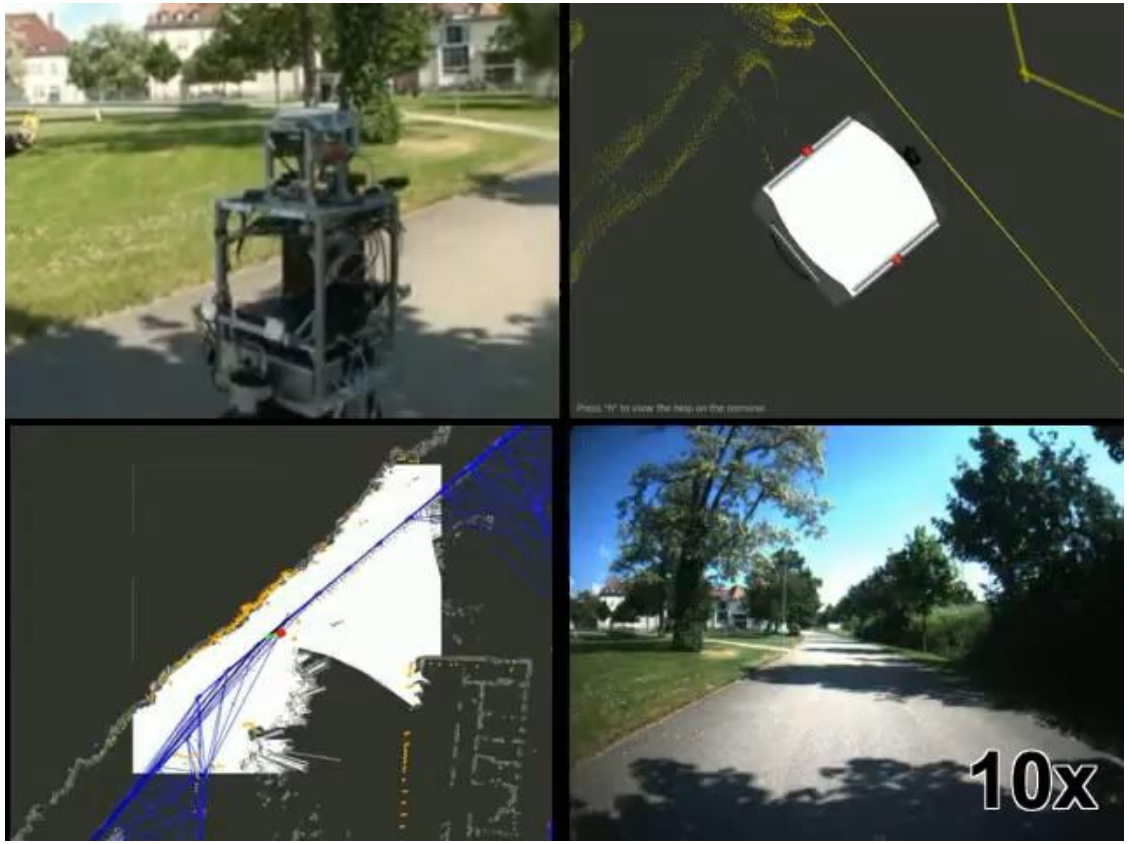








SLAM Showcase: Mapping Freiburg Campus



Courtesy of Cyril Stachniss

Problem Formulation

Input:

- The executed controls (e.g. steering, accel, decel) $u_{1:t} = \{u_1, \ldots, u_t\}$
- Observations made $z_{1:t} = \{z_1, \ldots, z_t\}$

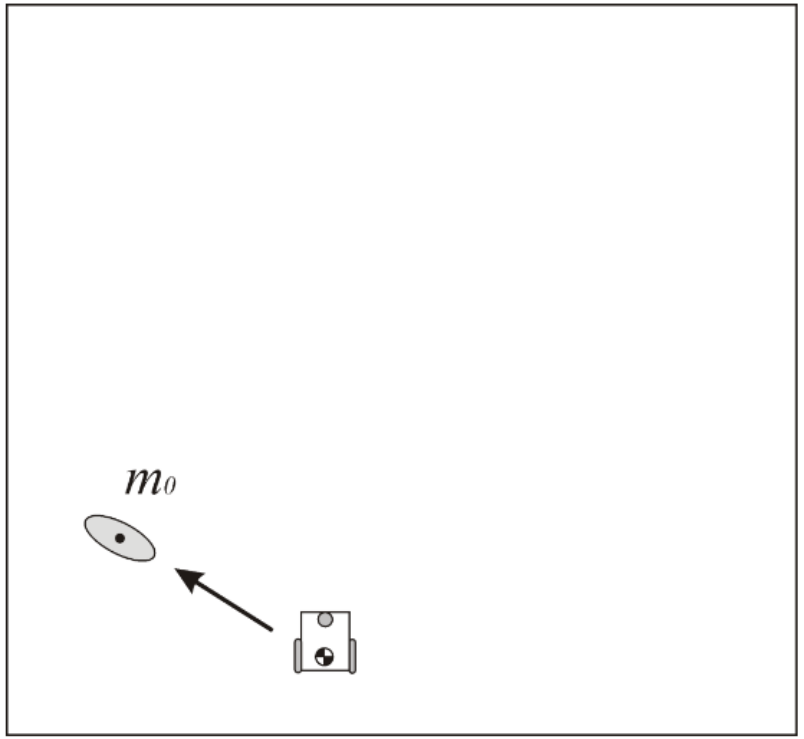
Wanted:

- Map of the environment m
- Path through the environment $x_{0:t} = \{x_0, \ldots, x_t\}$

Probabilistic formulation: $p(x_{0:t}, m | z_{1:t}, u_{1:t})$

Simultaneous Localization and Mapping (SLAM)

- Initial position perfectly known, i. e. $\boldsymbol{P}_{xx} = \boldsymbol{0}$
- Landmark position $\boldsymbol{m}_0$ sensed with uncertainty

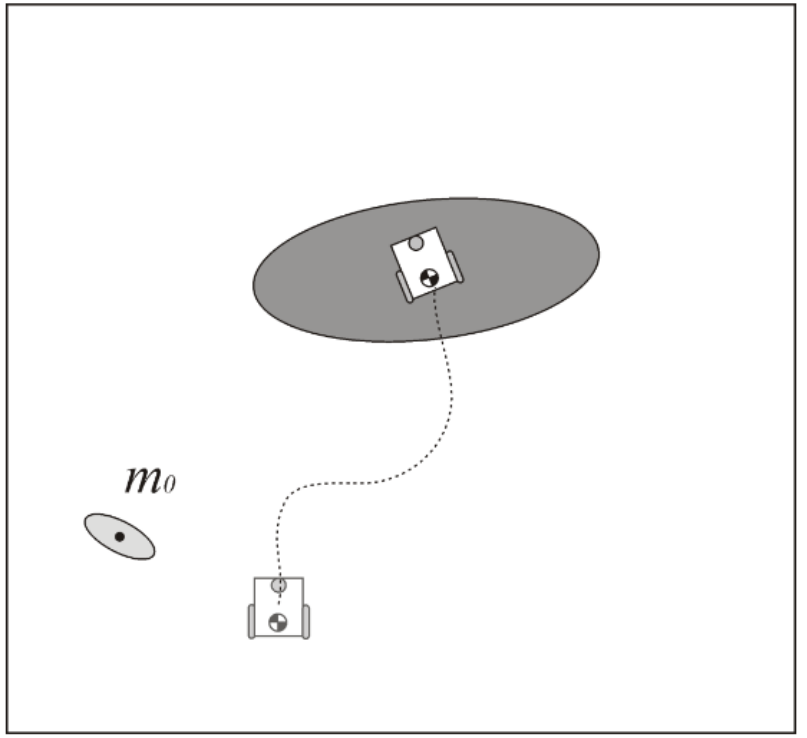


SLAM scenario, Figure borrowed from [2]

[2] https://www.cs.columbia.edu/~allen/F19/NOTES/slam_pka.pdf

Simultaneous Localization and Mapping (SLAM)

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- When the robot moves, the uncertainty of its state estimate increases

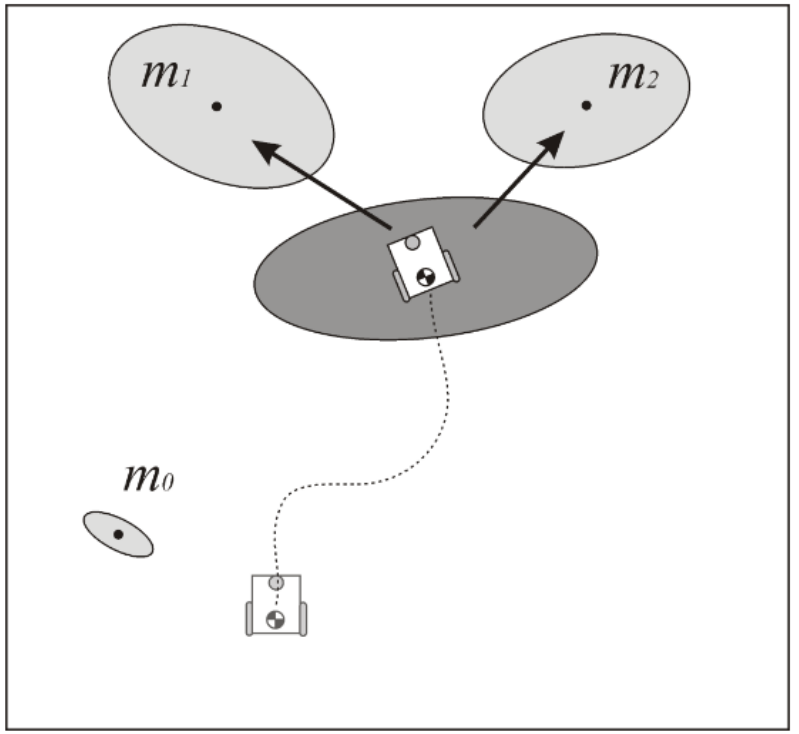


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Simultaneous Localization and Mapping (SLAM)

- Initial position perfectly known, i. e. $\boldsymbol{P}_{xx} = \boldsymbol{0}$
- Landmark position $\boldsymbol{m}_0$ sensed with uncertainty
- When the robot moves, the uncertainty of its state estimate increases
- Now, two new landmarks are sensed with an uncertainty determined by the own position uncertainty and the measurement uncertainty
  - Landmark and state estimate become correlated

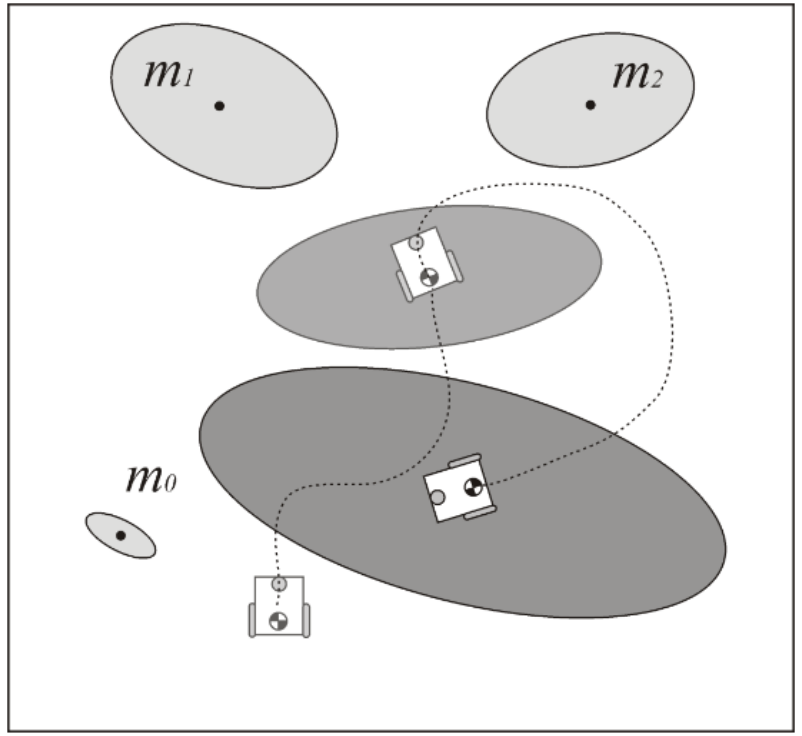


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  - Landmark and state estimate become correlated
- Robot moves again, state estimate uncertainty increases

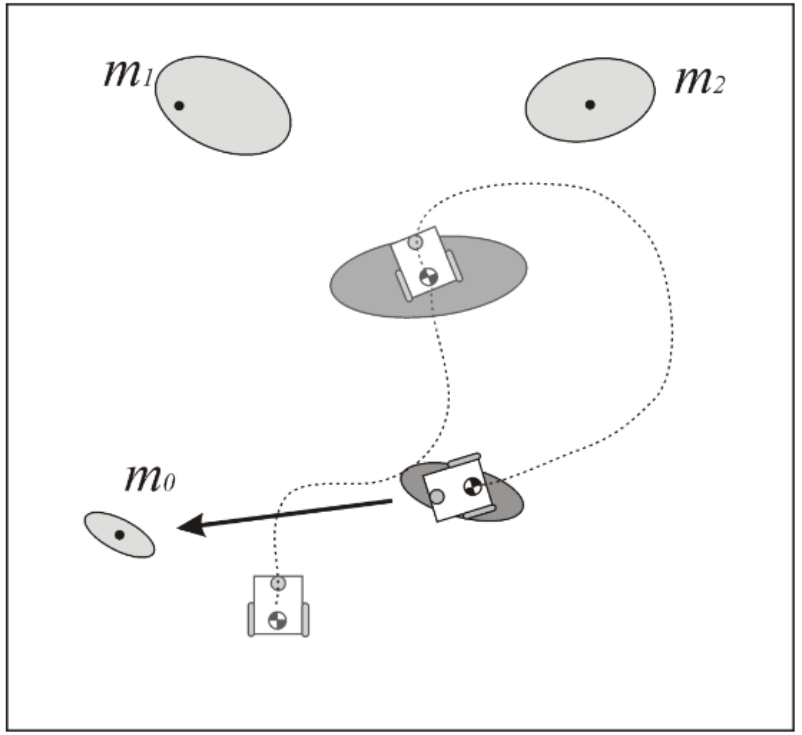


SLAM scenario, Figure borrowed from [2]

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Simultaneous Localization and Mapping (SLAM)

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- Landmark position $\boldsymbol{m}_0$ sensed with uncertainty
- When the robot moves, the uncertainty of its state estimate increases
- Now, two new landmarks are sensed with an uncertainty determined by the own position uncertainty and the measurement uncertainty
  - Landmark and state estimate become correlated
- Robot moves again, state estimate uncertainty increases
- Now the landmark m_0 is sensed again, which has a low uncertainty, this is referred to as loop-closure
  - Then the uncertainty of its state estimate is decreased
  - Simultaneously, all the uncertainties of the landmarks decrease

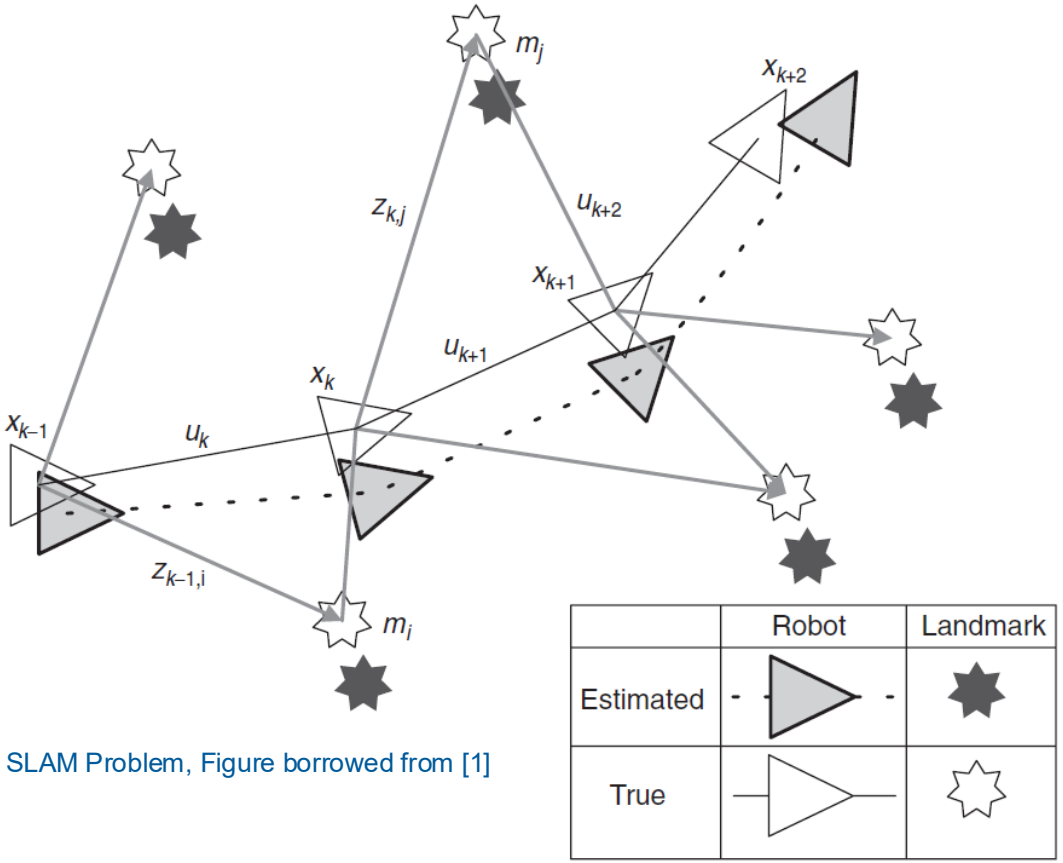


SLAM scenario, Figure borrowed from [2]

[2] https://www.cs.columbia.edu/~allen/F19/NOTES/slam_pka.pdf

Simultaneous Localization and Mapping (SLAM)

- Due to common localization error, the errors in landmark positions are highly correlated
  - Correlations increases with more observations
- Relative locations does not depend on the robot systems coordinate frame
  - Relative locations $\boldsymbol{m}_i - \boldsymbol{m}_j$ of landmarks may be known with high accuracy
  - Knowledge of relative positions improves with more observations
- Observations by the robotic system can be considered nearly independent measurements of relative locations

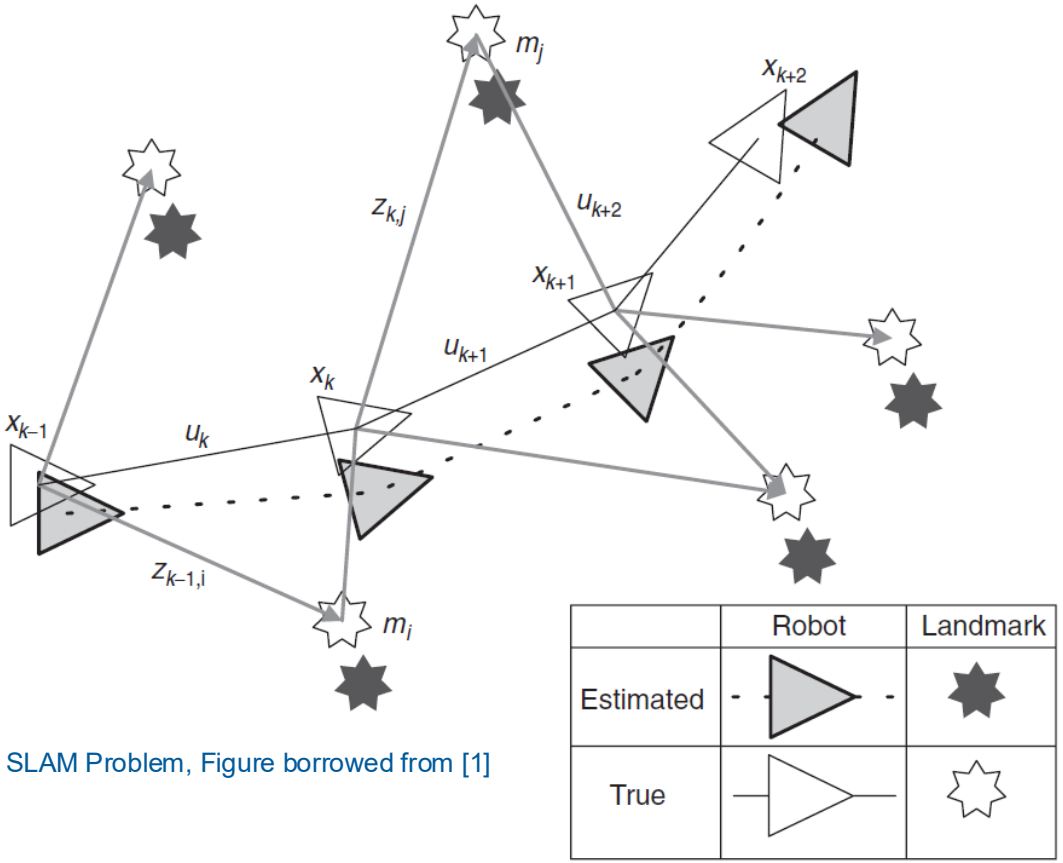


SLAM Problem, Figure borrowed from [1]

[1] Durrant-White et. al.: Simultaneous Localization and Mapping: Part 1, IEEE Robotics & Automation Magazine, 2006

Simultaneous Localization and Mapping (SLAM)

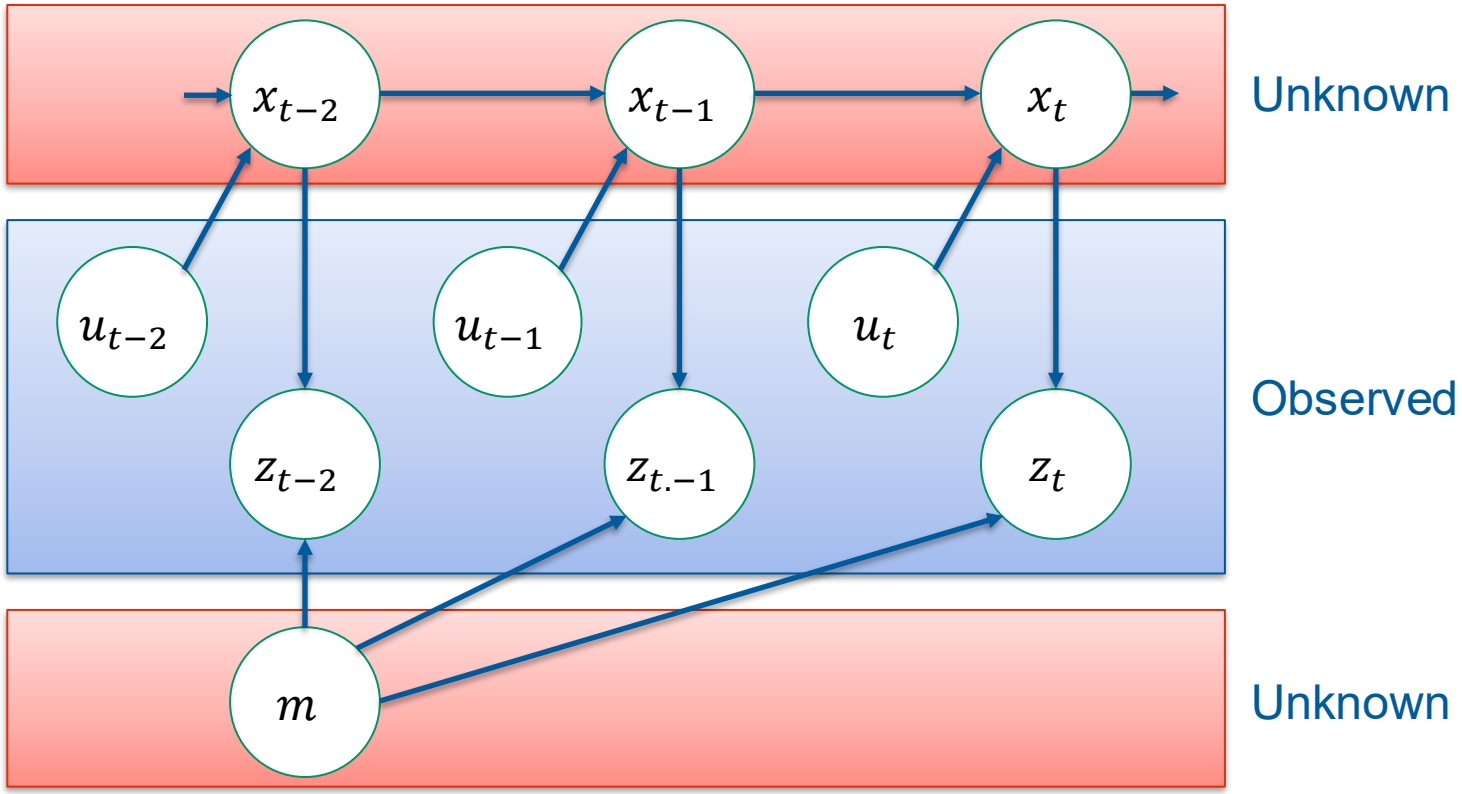
- A moving robot observes the same landmark $\boldsymbol{m}_j$ two times at $\boldsymbol{x}_k$ and $\boldsymbol{x}_{k+1}$
- At $\boldsymbol{x}_{k+1}$ with the new observation position and landmark are updated and due to the high correlations, this correction also affects $\boldsymbol{m}_i$



SLAM Problem, Figure borrowed from [1]

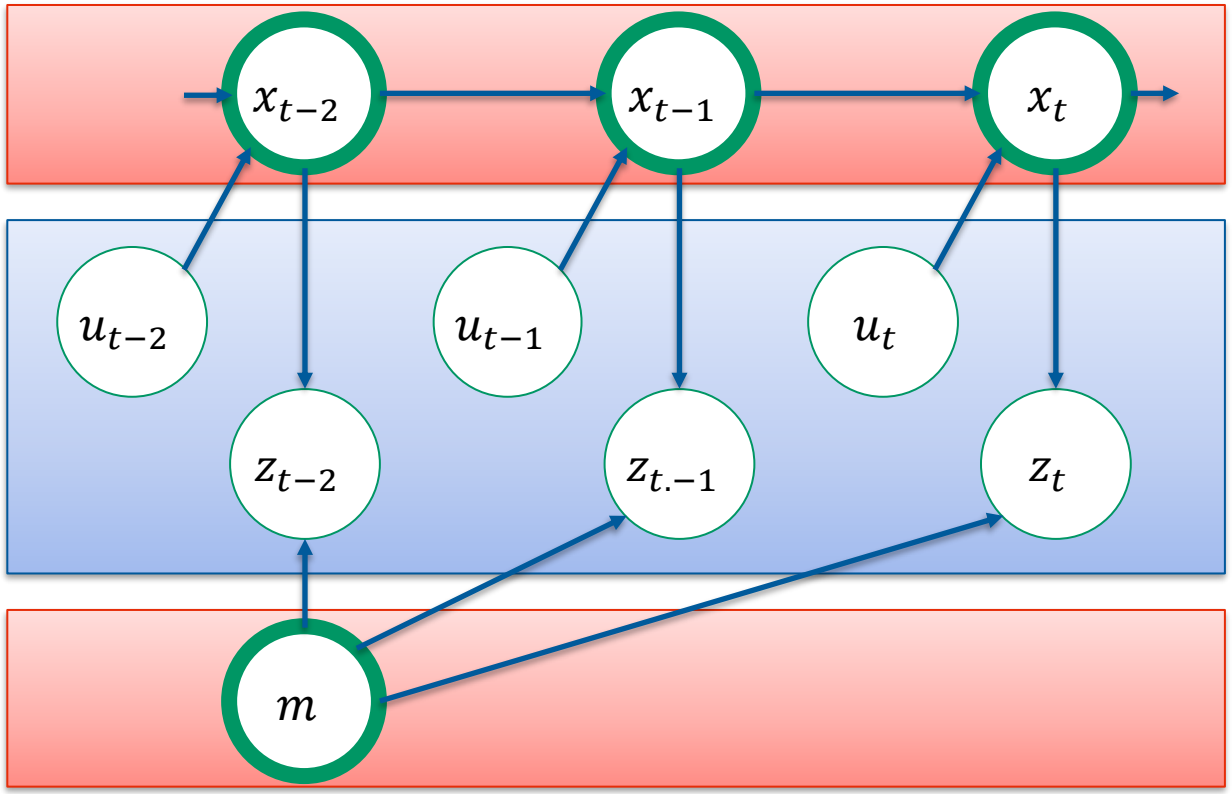
[1] Durrant-White et. al.: Simultaneous Localization and Mapping: Part 1, IEEE Robotics & Automation Magazine, 2006

Graphical Model



Picture: [1]

Graphical Model

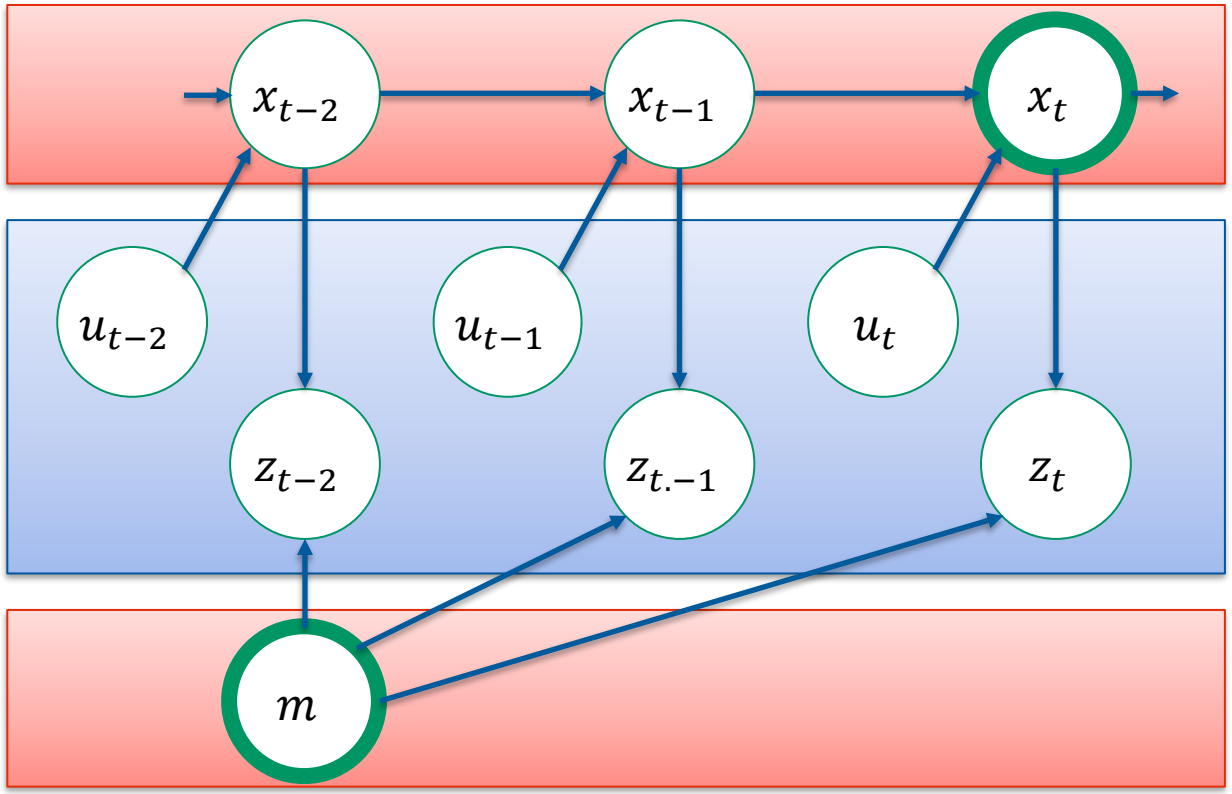


Full SLAM estimates the entire path:

$$p(x_{0:t}, m | z_{1:t}, u_{1:t})$$

Picture: [1]

Graphical Model



Picture: [1]

Full SLAM estimates the entire path:

$$p(x_{0:t}, m|z_{1:t}, u_{1:t})$$

Online SLAM only recovers the most recent pose:

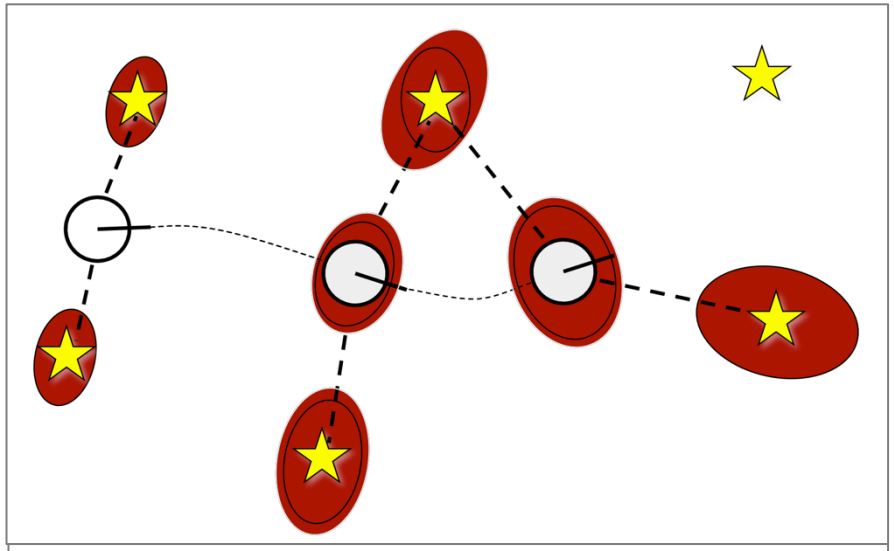
$$p(x_t, m|z_{1:t}, u_{1:t})$$

...by marginalizing out the previous poses:

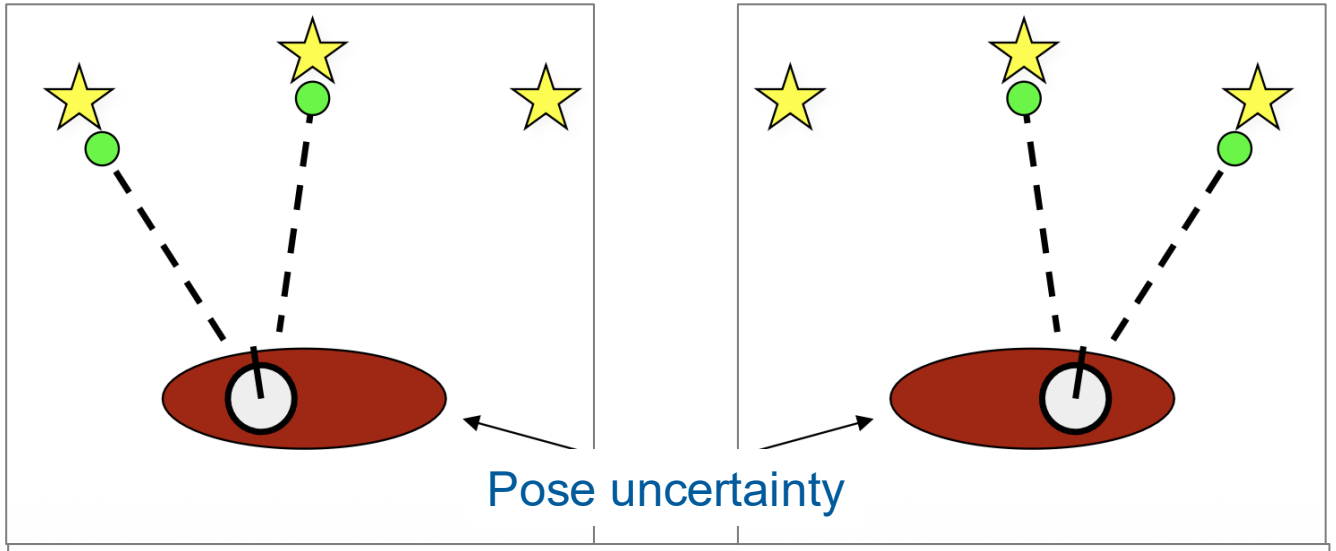
$$p(x_t, m|z_{1:t}, u_{1:t}) = \int \dots \int p(x_{0:t}, m|z_{1:t}, u_{1:t}) dx_{t-1} \dots dx_0$$

Integrals are typically solved recursively step by step.

SLAM is a Hard Problem



Robot path and map both unknown.
Map and pose estimates correlated.



Mapping between observations and map is unknown.
Picking wrong associations can cause the system to diverge.

SLAM Families

EKF SLAM

- Model entire problem with Kalman Filter.
- Idea: Unify map and robot state in one state vector / covariance matrix.
- Disadvantage: Very computationally intensive

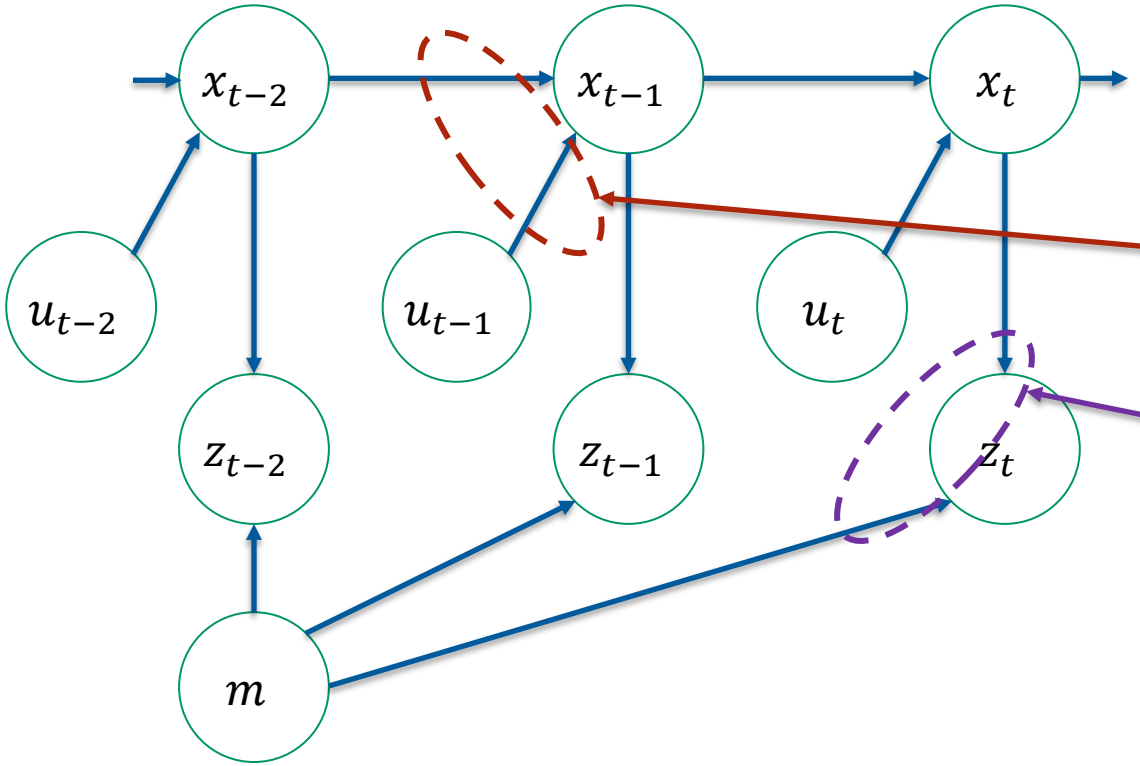
Particle Filter SLAM

- Model uncertainty only in the robot's pose
- One particle per hypothesis
- Each hypothesis creates its own map
- Prune whenever possible, leaving best map / particle combination

Least Squares / Graphs

- Model inputs and observations in a graph structure
- Optimize over graph with Least Squares methods

Motion and Observation Model



Like we have seen in the Kalman filter lectures, we want to use as much information as we can about our system.

The **Motion Model** captures our a priori idea of how our robot moves.

The **Observation Model** captures our idea of how (and how accurately) landmarks can be measured by our sensors.

Picture: [1]

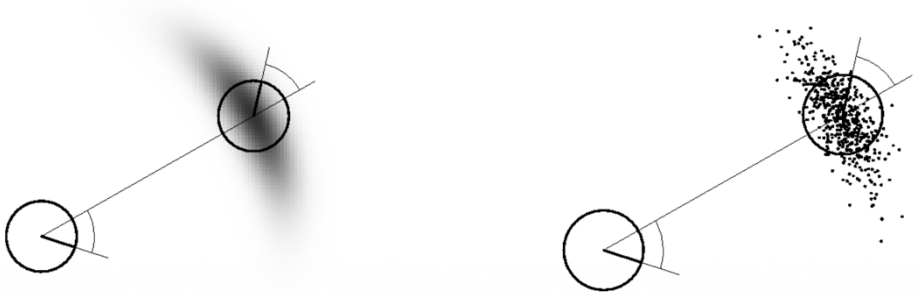
Motion Model

- **The motion model describes the relative motion of the robot.**

$$p(x_t|x_{t-1}, u_t)$$



Gaussian Motion Model

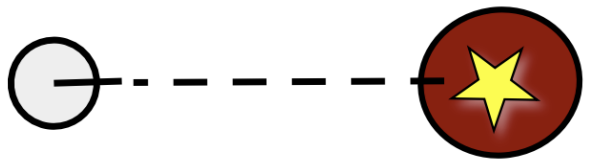


Non-Gaussian Motion Model

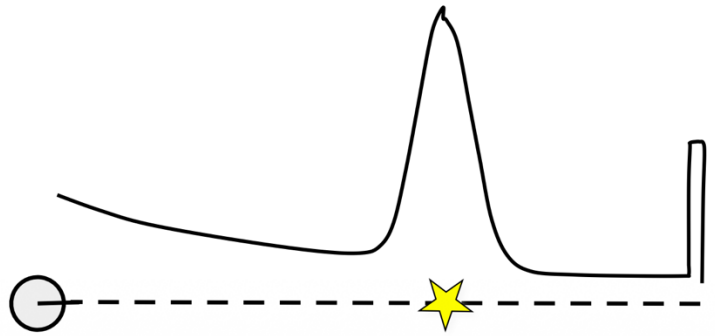
Observation Model

- **The observation or sensor model relates landmark measurements with the robot's pose.**

$$p(z_t|x_t)$$



Gaussian Observation Model



Non-Gaussian Observation Model

Sensor Networks and Sensor Fusion

- **Previous lectures:**
  - Mathematical details of Particle Filters
  - Joint Probability Data Association Filters
  - JPDAs and Particle Filters

- **This week:**
  - Simultaneous Localization and Mapping
    - Introduction
    - EKF SLAM
    - Loop Closure
    - Particle Filter / Rao-Blackwellized Particle Filter SLAM
    - "Hidden Robot" problem
- **After the Christmas Break:**
  - Bit more SLAM
  - (DNN-based) early fusion approaches.



Merry Robot Christmas!

Literature / Links

1. **S.Thrun &al., *Probabilistic Robotics*, https://robots.stanford.edu/probabilistic-robotics/**



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Thank you very much!